PROJECT PROPOSAL · PRESENTATION

Off-Grid LoRa-Based Self-Healing Mesh Communication Infrastructure

with an Autonomous Rescue Rover for Emergency Search & Rescue Operations

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WHY WE CHOSE THIS PROJECT

Motivation

- Disasters destroy the very infrastructure rescue depends on — towers, power lines and internet links fail exactly when coordination matters most.
- Most rescue communication tools still assume a working cellular or internet backbone, so they fail in the worst-hit zones.
- The first 24–72 hours are the highest-survival window; minutes lost to poor coordination cost lives.
- Rescuers are forced to physically enter collapsed or hazardous structures just to check whether anyone is inside.
- Bangladesh and similar flood- and earthquake-prone regions need a solution that is low-cost, portable and locally repairable.

Faster Coordination

Teams stay linked without any network, so decisions happen in seconds, not hours.

Safer Operations

The rover enters what humans should not, reducing risk to rescue personnel.

Accessible & Affordable

Off-the-shelf parts keep the system deployable by local authorities and volunteers.

PROBLEM STATEMENT

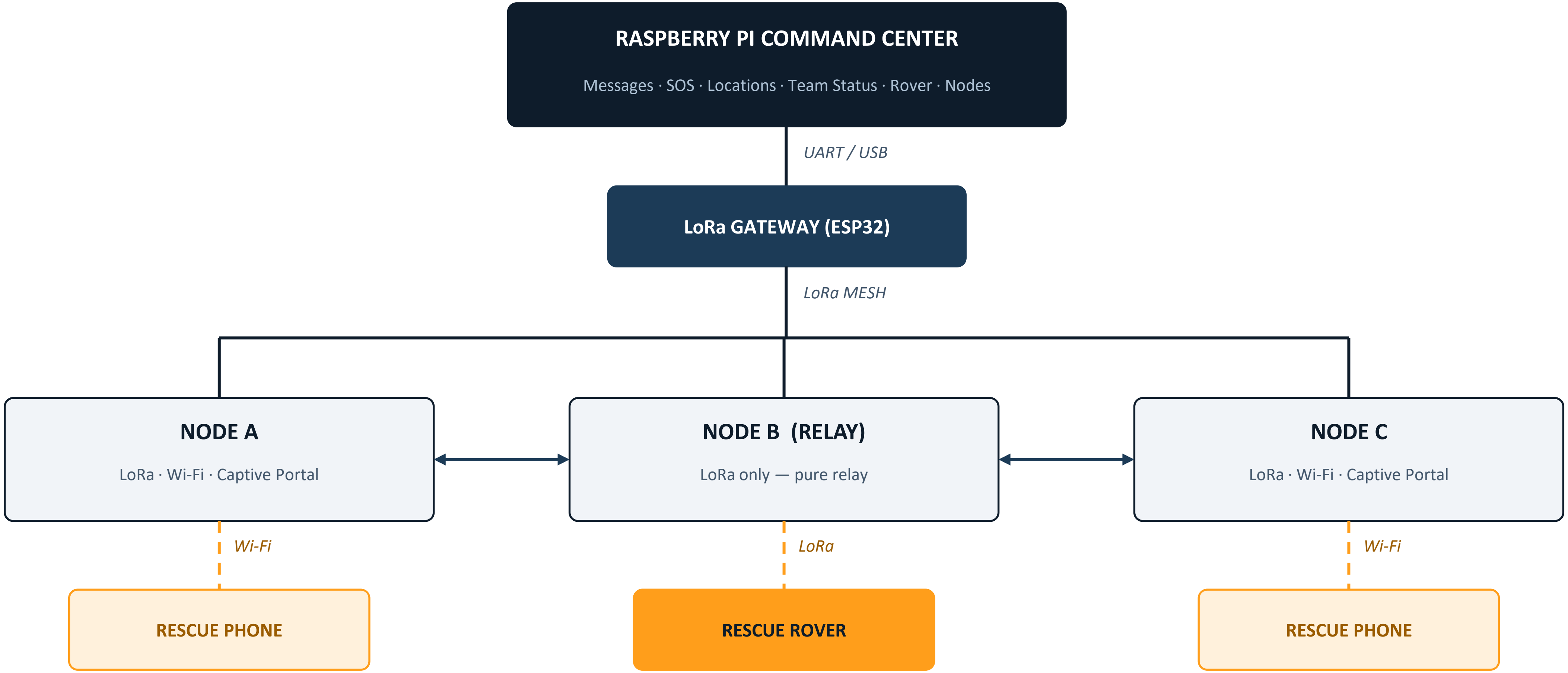
Introduction — The Problems We Solve

1. No reliable communication between rescue personnel once towers go down
2. Difficulty coordinating multiple teams across a wide disaster zone
3. Limited or zero coverage in remote and badly-affected areas
4. No easy way to share the live location of each rescue team
5. Emergency alerts cannot be raised or escalated quickly
6. Communication collapses whenever an intermediate node fails
7. Dedicated GPS is unreliable indoors, under roofs and beneath debris
8. Unsafe for humans to enter collapsed structures and narrow passages

CORE PROBLEM

A decentralised, off-grid infrastructure is needed that keeps rescuers connected AND can explore places humans cannot safely reach.

Project Diagram



Rover roles: Autonomous Movement · Obstacle Avoidance · Mobile LoRa Relay

SYSTEM OVERVIEW

Proposed Solution

01

LoRa Mesh Nodes

Decentralised long-range radio backbone with multi-hop forwarding and automatic routing.

02

Wi-Fi Captive Portal

Browser-based rescue interface — no dedicated mobile application to install.

03

Autonomous Rover

Explores hazardous zones and doubles as a repositionable mobile LoRa relay.

04

Pi Command Center

Central monitoring of messages, alerts, locations, team status and rover health.

MESH CAPABILITIES

- Long-Range
- Multi-Hop
- Auto Discovery
- Reliable Delivery
- Heartbeat
- Self-Healing

All four components operate as one integrated system — communication, mobile access, exploration and monitoring — with zero dependence on cellular or internet infrastructure.

Key Features — LoRa Mesh Network

01

Long-Range Communication

LoRa radio links nodes over long distances with no cellular or internet dependency.

02

Multi-Hop Forwarding

Messages relay through intermediate nodes: Node A → Node B → Node C.

03

Automatic Node Discovery

Nodes detect and register nearby neighbours without manual configuration.

04

Reliable Messaging

Message IDs, ACK responses, retries, timeout detection and duplicate filtering.

05

Heartbeat Monitoring

Periodic beacons confirm which nodes are alive and reachable in real time.

06

Self-Healing Routing

If a node drops out, the network automatically re-routes through an alternative path.

Key Features — Wi-Fi Captive Portal

RESCUE PORTAL

Node A · Team 04 · Connected

SEND SOS

Share My Location

Send Short Message

Quick Rescue Report

Update Team Status

no app install required

Wi-Fi Access Point

Every rescue node broadcasts its own Wi-Fi for nearby responders.

Captive Portal

The interface opens automatically in any browser — nothing to install.

User & Team Identity

Responders register a User ID, Team ID and rescue unit.

Mobile GPS Sharing

Phone sends latitude, longitude, accuracy, timestamp and source.

SOS Emergency Alert

Raised from the portal or from the physical button on the node.

Short Messaging

Send to the command centre, another team, or broadcast to all nodes.

Quick Rescue Reports

One-tap presets: Victim Found, Medical Help, Area Blocked, Danger.

Team Status Updates

Available · Searching · Victim Found · Need Assistance · Emergency.

Key Features — Rover & Command Center

AUTONOMOUS RESCUE ROVER

- Autonomous exploration of dangerous, narrow or inaccessible areas
- Ultrasonic obstacle detection and avoidance around walls, debris and blocked paths
- Full LoRa mesh membership — sends status and location, receives commands
- Mobile LoRa relay: physically repositions to bridge separated nodes
- Reports rover ID, movement, obstacle state, battery level and link quality

RASPBERRY PI COMMAND CENTER

- Network monitoring — live view of every node in the mesh
- Message monitoring — incoming rescue traffic displayed and stored
- SOS monitoring — emergency alerts surfaced prominently
- Location monitoring — positions from phones and GPS-enabled nodes
- Team status board — operational state of every rescue unit
- Rover monitoring and local data logging for post-operation review

KEY IDEA: the rover is not just a robot — by relaying LoRa traffic on the move, it becomes a living part of the communication infrastructure itself.

Apparatus I — Rescue Communication Node

1

ESP32 Development Board

Main controller, Wi-Fi access point, captive portal server and mesh logic.

2

LoRa Transceiver

SX1278 / SX1276 — long-range links, mesh routing and message forwarding.

3

GPS Module

Neo-6M / Neo-M8N — physical position whenever satellite signal is available.

4

0.96" I2C OLED Display

Shows node ID, LoRa state, GPS fix, SOS alerts and network information.

5

SOS Push Button

Hardware emergency trigger for when a phone is unavailable or unusable.

6

Battery / Power Bank

Portable power so nodes can be dropped anywhere without mains supply.

Apparatus II — Rover & Command Center

AUTONOMOUS ROVER

- ESP32 development board
- LoRa module (SX1278 / SX1276)
- Rover chassis — 2WD or 4WD
- DC motors
- Motor driver (L298N / TB6612FNG)
- Ultrasonic sensor HC-SR04
- GPS module (Neo-6M / Neo-M8N)
- Battery system for motors + logic

GATEWAY & COMMAND CENTER

- ESP32 LoRa gateway node
- LoRa module (mesh bridge)
- Raspberry Pi 4 or Raspberry Pi 5
- USB / UART link to the gateway
- Local storage for message logging
- Display for the monitoring dashboard

USER DEVICE

- Any Wi-Fi smartphone
- Connects to node Wi-Fi
- Opens the captive portal
- Supplies browser GPS location
- Sends messages and SOS alerts
- Submits rescue reports
- Updates rescue team status

Software & Technologies

1

Embedded

- Arduino IDE
- PlatformIO
- C / C++

2

ESP32 Layer

- Wi-Fi access point
- HTTP web server
- Captive portal
- LoRa libraries

3

Raspberry Pi

- Python
- UART / serial I/O
- Local data storage

4

Web Interface

- HTML
- CSS
- JavaScript

5

Mobile Location

- Browser Geolocation API
- Hybrid GPS fallback

WHY THIS STACK

Open-source, lightweight and widely supported — keeping the system cheap to build, easy to debug in the field and simple for another team to maintain.

Expected Outcome & Significance

DEMONSTRATED CAPABILITIES

Long-range LoRa link	Multi-hop mesh routing	Automatic node discovery
Reliable ACK messaging	Heartbeat monitoring	Self-healing re-routing
Wi-Fi captive portal	Mobile GPS sharing	SOS alerts & reports
Team status updates	Autonomous rover run	Mobile LoRa relay

SIGNIFICANCE

- Zero dependency on cellular or internet infrastructure
- Coverage grows simply by adding more nodes
- Network survives individual node failures
- Works with any smartphone — no app to install
- Hazardous areas explored without risking rescuers
- Rover extends coverage dynamically as a mobile relay
- One central, low-cost, portable coordination point

Conclusion

An integrated off-grid mesh network, browser-based rescue portal, autonomous rover and central command post — built to work when nothing else does.

FUTURE SCOPE

- Camera integration
- Thermal victim detection
- Gas & smoke sensors
- AI-based victim detection
- Offline maps
- Advanced navigation
- Drone integration
- Encrypted communication
- Solar-powered nodes
- Environmental sensing
- Larger-scale mesh
- Field trials with rescuers

THANK YOU !